

Ainesh Sootha

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Experience

Amazon – Ads AI

Dec.2025 – Present

Software Engineer (ML)

New York, NY

- Building a globally distributed production LLM inference platform for Amazon Ads using Kubernetes, managing fleet orchestration, autoscaling, and multi-region deployment across AWS infrastructure.
- Engineering performance optimizations for production-scale LLM inference including disaggregated inference, distributed KV caching, suffix decoding, and speculative decoding—translating novel research into scalable solutions.
- Integrating and maintaining inference framework integrations (vLLM, tensorRT-LLM, NVIDIA Dynamo) and building deployment pipelines, monitoring infrastructure, and operational tooling, in collaboration with Nvidia.
- Wrote multi-agent orchestration frameworks to automate replica management, inference optimizations, configuration search space exploration, balancing a global capacity pool.

Amazon Web Services - Bedrock AI

Apr.2024 – Dec.2025

Software Engineer (ML)

Seattle, WA

Bedrock Generative AI

- Engineered performance optimizations for production-scale open source LLMs (Llama, Deepseek, Mistral) on AWS Bedrock, translating novel research into scalable solutions through collaboration with research scientists.
- Experimented with a wide range of novel techniques including distributed caching, distributed inference, and speculative decoding to achieve measurable performance improvements. This involved a lot of rapid prototyping and extensive benchmarks (and automation thereof).
- Architected and built Custom Model Import, a serverless multi-tenant platform serving 3,000+ LLM endpoints, enabling seamless model deployment, serving, and monitoring for AWS Bedrock customers.
- Partnered with a diverse set of external customers ranging from teams of MLEs to research groups, delivering hyper-optimized model deployments (on AWS Bedrock Custom Model Import) for specific performance requirements.

Amazon

Jul.2023 – Apr.2024

Software Engineer

Seattle, WA

Just Walk Out Technology – Auth Orchestration

- Worked on low-latency software for orchestration of customer authentication and authorization. (Java, Kotlin)

Amazon

May.2022 – Aug.2022

Software Engineer Intern

Seattle, WA

Just Walk Out Technology – Customer Identification & Tracking

- Built a trajectory visualization web application that accurately mapped customer journeys across stores.

State Street Corp

Jun.2021 – Aug.2021

Software Engineer Intern

New York, NY

- Implemented a custom PDF generation service for **BestX** TCA dynamic web reports using chromium and Node.

Projects and other work (github.com/aineshsootha)

Realistic Ocean Simulation - GPU-parallelized generation of realistic Ocean Waves (Tessendorf) (C++, OpenGL)

Education & Skills

Purdue University

2019 – 2023

BS, Computer Engineering + Minor in Philosophy

West Lafayette, IN

- **Relevant Courses:** Data Structures, Computer Graphics, OS, OOP in C++, Compilers, Networks, Linear Algebra